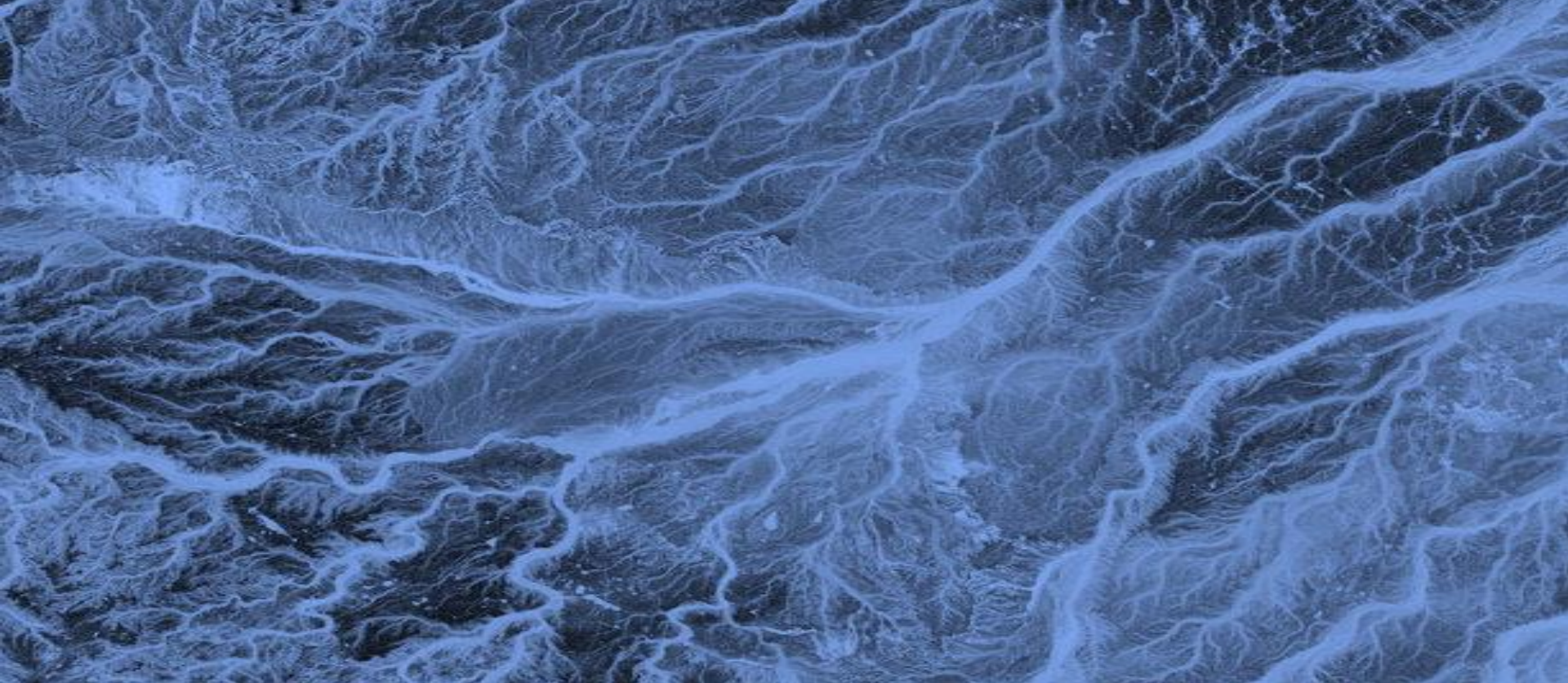




Modeling Stream Network Environments

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Overview

- **Conceptualization of stream systems**
- **Key Concepts**
- **Challenges of Modeling Streams Data**
- **Taxonomy of Network Analyses**



Douglas River, Northern Territory, Australia. Photo courtesy of Mark Kennard.

NCEAS Spatial Stats for Streams Working Group

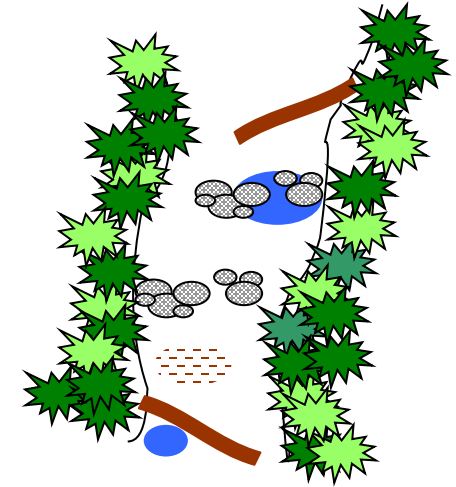


Peterson et al. (2013)

Conceptualization of Stream Systems

Highly connected, directed dendritic networks

- Dendritic ecological networks (Grant et al. 2007)
- **Seminal papers:** Schlosser & Angermeier 1995; Fisher 1997; Ward 1997,1998; Fausch et al. 2002; Power & Dietrich 2002; Wiens 1999, 2002; Allan 2004; Benda et al. 2004; Fisher et al. 2004
- **Recent review:** Torgersen et al. 2022



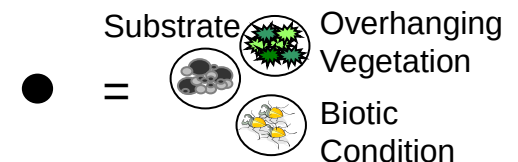
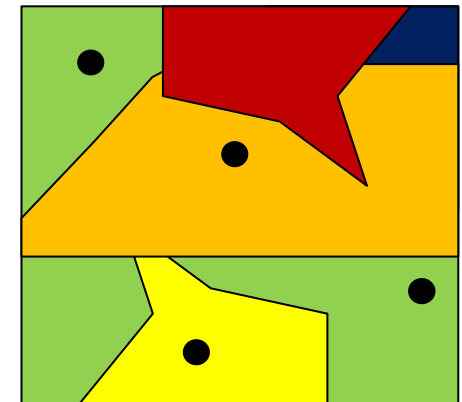
Studies often undertaken at two scales

1. Local studies (< 500m)

- Abundance & biotic interactions

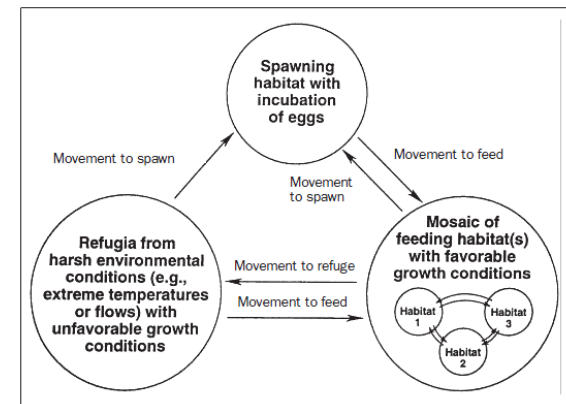
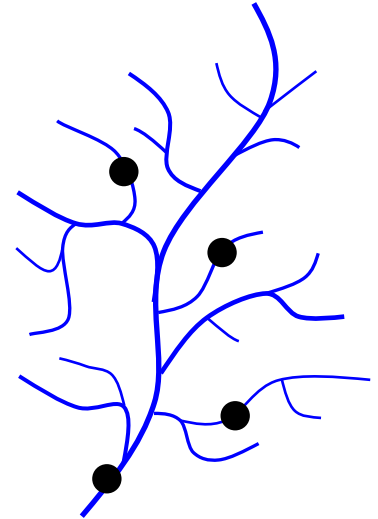
2. Macro-scale studies (> 100km)

- Coarse catchment or network averages used to make inferences about
 - species distributions
 - evolutionary processes
 - climate change
- Locations are treated as independent



Conceptualization of Stream Systems

- Highly connected, directed dendritic networks
 - Dendritic ecological networks Grant et al. 2007)
 - **Seminal papers:** Schlosser & Angermeier 1995; Fisher 1997; Ward 1997,1998; Fausch et al. 2002; Power & Dietrich 2002; Wiens 1999, 2002; Allan 2004; Benda et al. 2004; Fisher et al. 2004
 - **Recent review:** Torgersen et al. 2022
- **Key physicochemical biological processes operate at intermediate scales (1-100 km)**
 - e.g. Metapopulation dynamics and disturbance regimes



Schlosser's dynamic landscape model of stream fish life history (Figure 2 from Schlosser and Angermeier 1995).

Conceptualization of Stream Systems

Network Scale

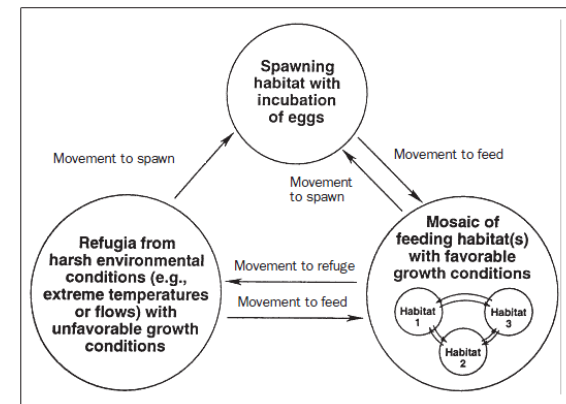
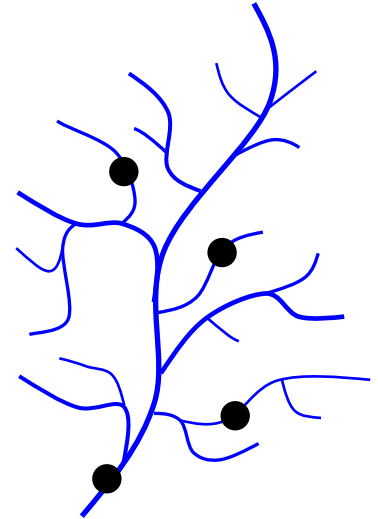
- Intermediate scale (1-100km)
- Investigate relationships between a set of components or locations, rather than treating discrete locations independently

Scientific Need

- To better understand key physicochemical and biological processes

Management Need

- To prioritize conservation actions



Schlosser's dynamic landscape model of stream fish life history (Figure 2 from Schlosser and Angermeier 1995).

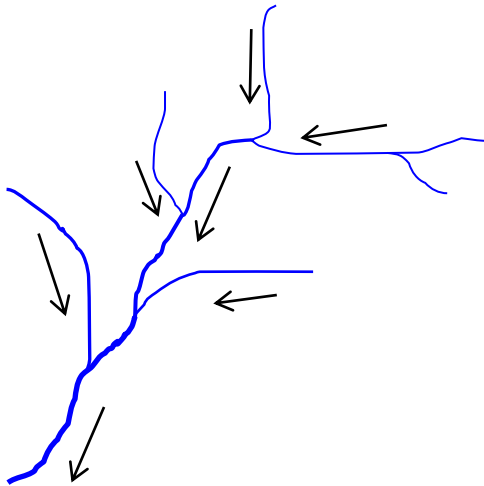
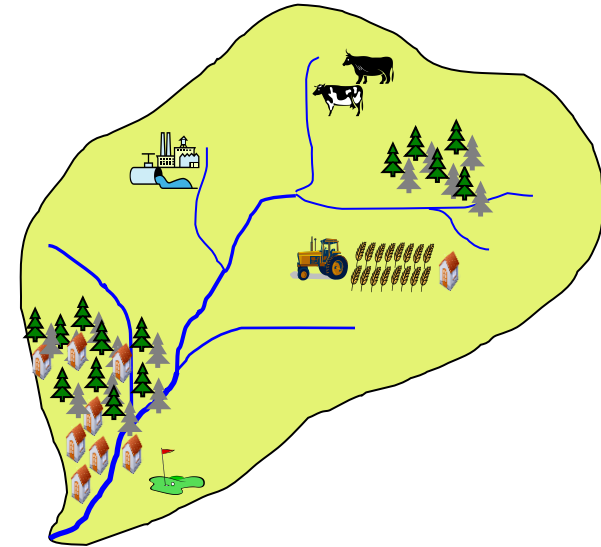
Key Concepts

Highly Connected Systems

- Spatially & temporally dynamic property

Lateral connectivity

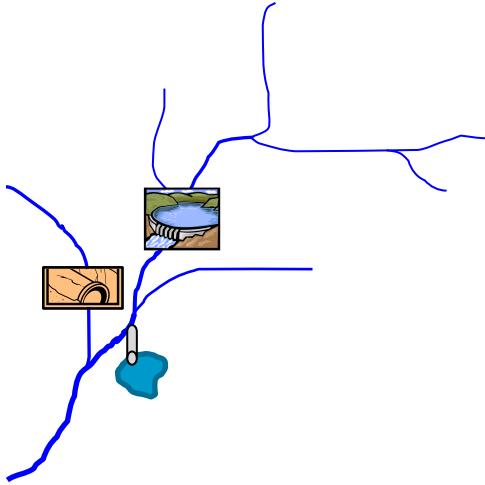
- Watershed-Floodplain-Channel
- Watershed & floodplain provide inputs of water, material, nutrients, organisms
- Networks are embedded in the terrestrial environment



Longitudinal connectivity

- Upstream-downstream
- In-stream processes alter form, concentration, and/or distribution of material & nutrients
- Water, material, & nutrients routed downstream

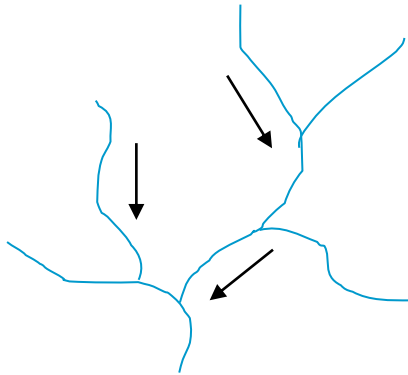
Key Concepts



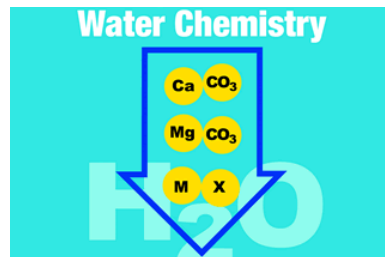
Networks form semi-restrictive corridors

- Limit movement of water, materials, & some organisms at various life stages
- Dams, water diversions, and culverts may result in longitudinal fragmentation
- Easily fragmented b/c few redundant pathways

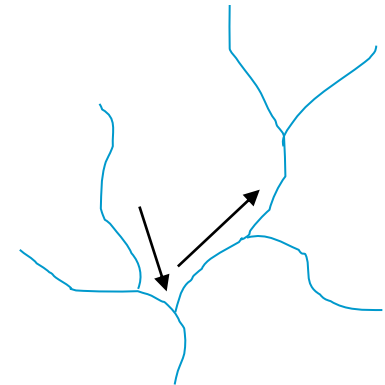
Directional Flow



Passive movement



(photo: Kluane National Park and Reserve)



Active movement

Key Concepts

Physical Network Structure Matters

- Dispersal of organisms
- Delivery of flows, effects of disturbance, availability of refugia
- Morphological heterogeneity at confluences controlled by size ratios of confluent tributaries
 - Provides inputs of water, material, & organisms
 - Creates heterogeneous habitat “patches”
 - Increases biodiversity

Position within network

- Food web structure and interactions with terrestrial environment
- Genetic diversity
- Species assemblages

Structure of river networks

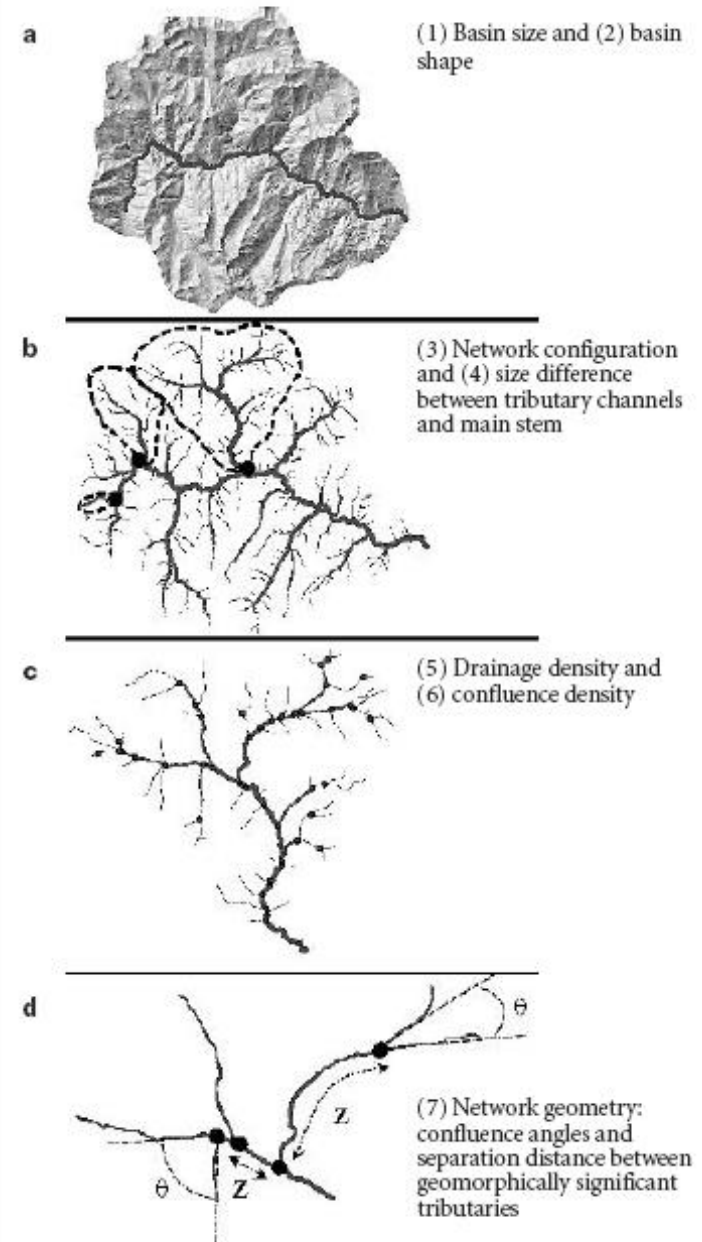
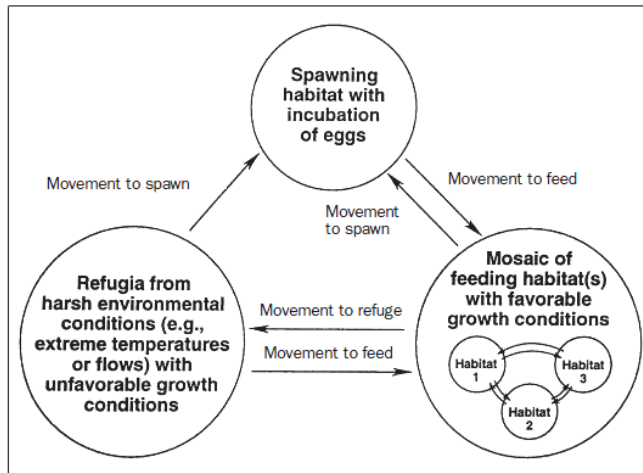


Figure 1 from Benda and others (2004). Seven structural indexes of river networks.

Key Concepts

Mobile organisms respond to **spatial arrangement of habitat** within or along the network, e.g. Fish and amphibians

- Temporally variable



Schlosser's dynamic landscape model of stream fish life history (Figure 1 from Schlosser and Angermeier 1995)

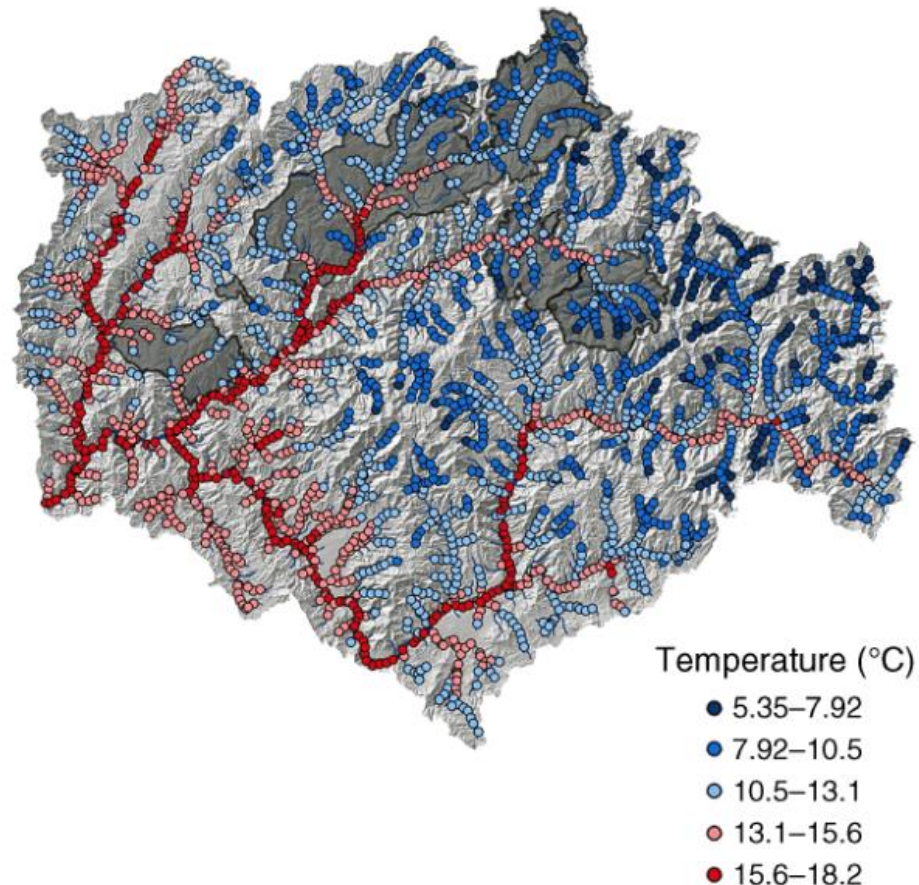


Figure 5a from Isaak et al. (2010)

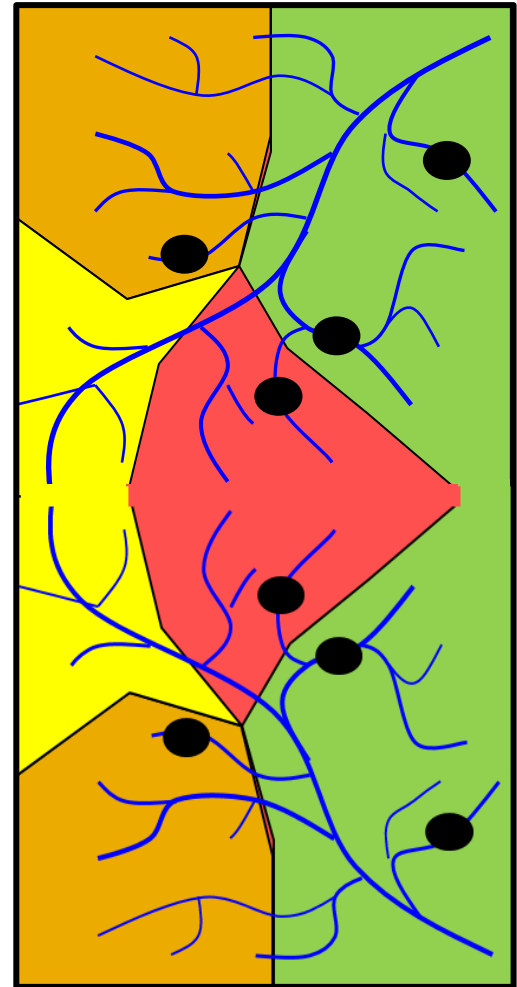
Key Concepts

- **Complex processes and landscape-scale patterns**
 - Composition and spatial arrangement of the elements
 - Species abundance
 - Nutrient concentrations
 - Intermittent versus perennial streams
 - Represent heterogeneity or variability
- **Produced by**
 - Biotic and abiotic conditions
 - Unique spatial characteristics of the stream
 - Ecological processes: hydrologic, physical, chemical and biotic
 - Past conditions
 - Disturbance
- **Multi-scale processes & interactions within and between aquatic and terrestrial environments produce complex, multi-scale patterns**

Challenges of Modeling Streams Data

Fundamental Stream Characteristics

- Dendritic network structure
- Connectivity
- Directional flow
- Spatio-temporal variability of habitat and flow
- Networks are embedded in the terrestrial environment



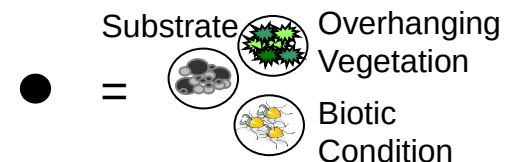
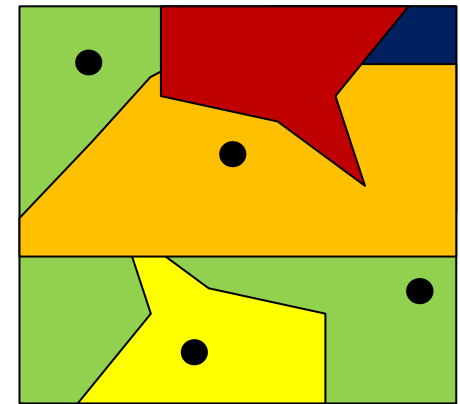
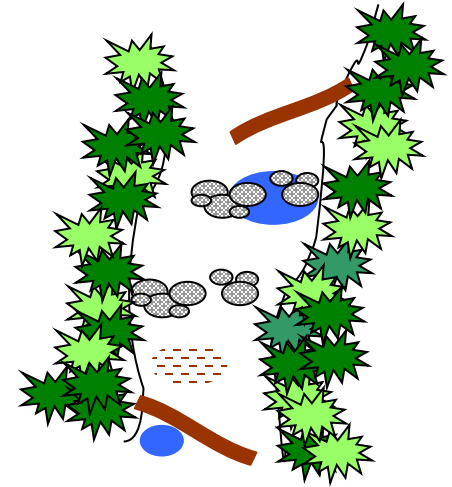
Challenges of Modeling Streams Data

Models that

- Ignore spatial relationships altogether
- Ignore network relationships

Create a mismatch between analytical approaches and our conceptualization of the system

- Limits our understanding of how streams function
- Weakens our ability to make accurate and unbiased predictions
- Reduces the effectiveness of management actions



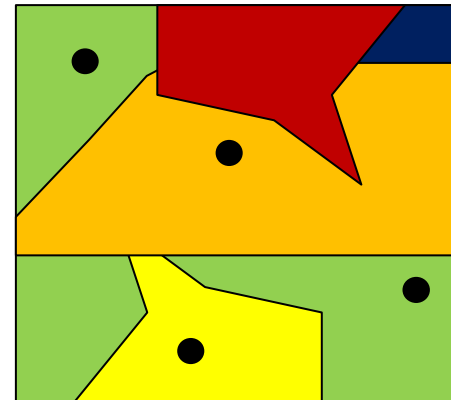
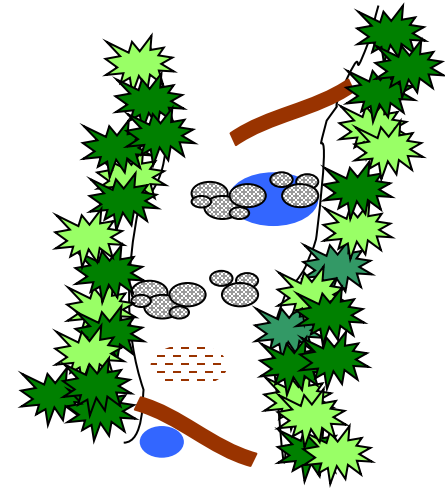
Challenges of Modeling Streams Data

Fundamental Network Characteristics

- Dendritic network structure
- Connectivity
- Directional flow
- Spatio-temporal variability of habitat and flow
- Networks are embedded in the terrestrial environment

Why are most studies focused on the local and macro scales?

- Ignore fundamental network characteristics

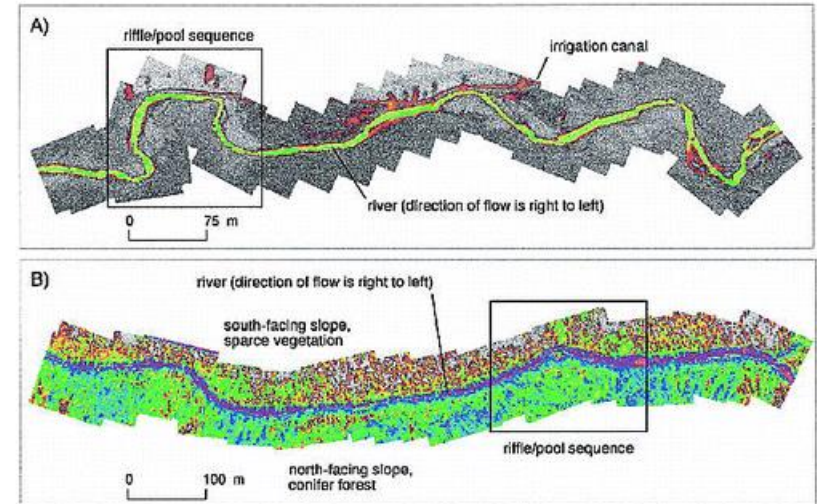


Challenges of Modeling Streams Data

Lack of data - spatially dense or continuous data is usually \$\$\$\$

Example: Torgersen et al. 1999

- Investigated the relationship between
 - Spring chinook salmon distribution
 - Fine-scale patterns of stream temperature and habitat
- Temperature data- FLIR
- Fish data- snorkeling 1km intervals in the upper sections of the Middle Fork and North Fork of the John Day River, Oregon
- Leads to technical and financial tradeoffs (Torgersen et al. 2022)



Reach-level patterns of stream temperature.
Figure 3 from Torgersen et al. 1999



C. Torgersen in action

Challenges of Modeling Streams Data

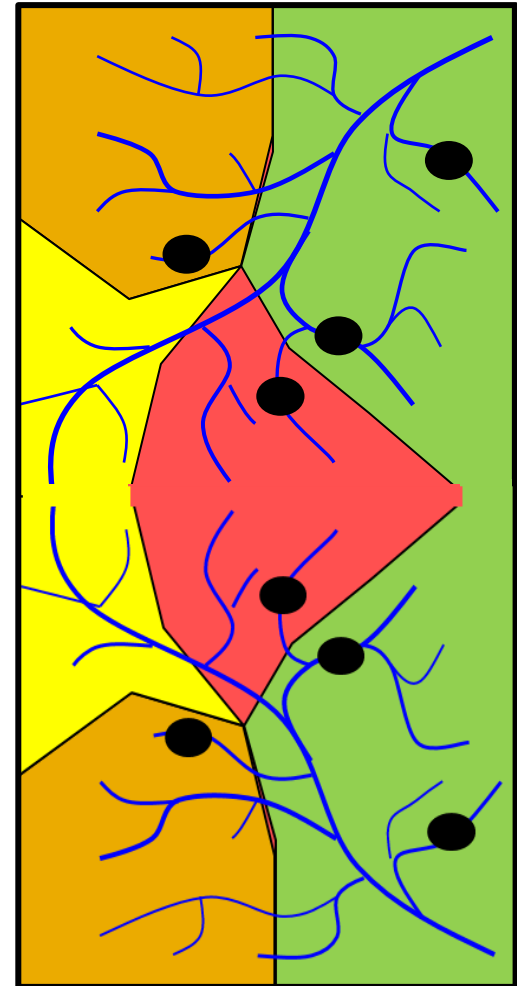
Unfamiliarity with suitable methods

- Graph-theoretic-based approaches (GT)
- Metapopulation models
- Spatial statistical approaches

All account for structure, connectivity, & directionality to some degree

Example: Schick & Lindley (2007)

- Large storage dams blocked access to spawning habitat, leading to declines in abundance and distribution of spring chinook salmon populations in CA Central Valley
- Used GT and standard graph metrics to understand changes in connectivity
- Prioritize dam removal and restoration activities



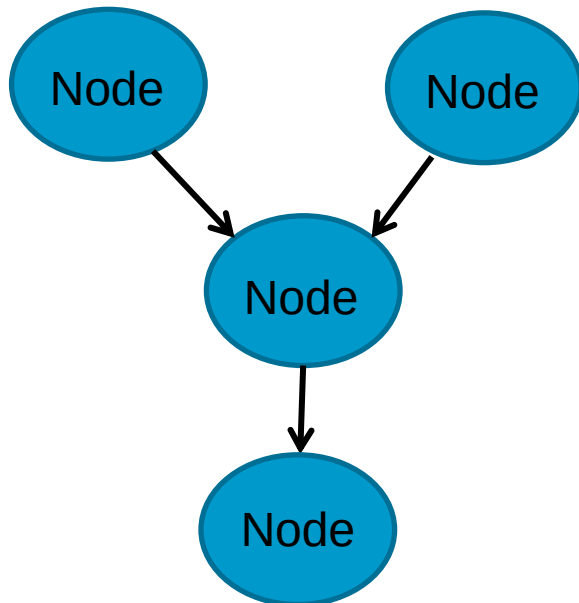
Challenges of Modeling Streams Data

Application to stream environments may not be intuitive

Graph-theoretic and Metapopulation models:

- Nodes represent habitat patches or populations
- Edges represent functional relationships: rate of dispersal

In streams processes occur *ON* the network



Challenges of Modeling Streams Data

Streams data have a dual spatial representation

- Streams are embedded in the terrestrial environment
- Spatial representation in network & 2-D space
 - Longitudinal & lateral connectivity
 - Climatic, topographic, geologic gradients

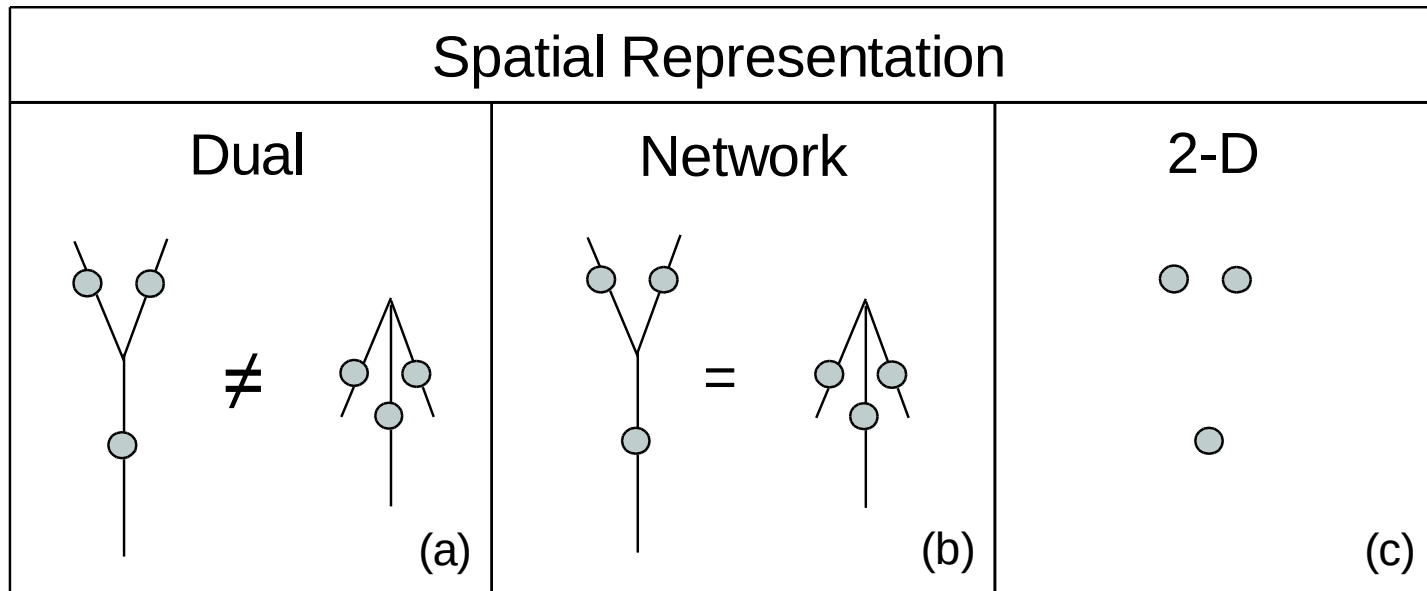
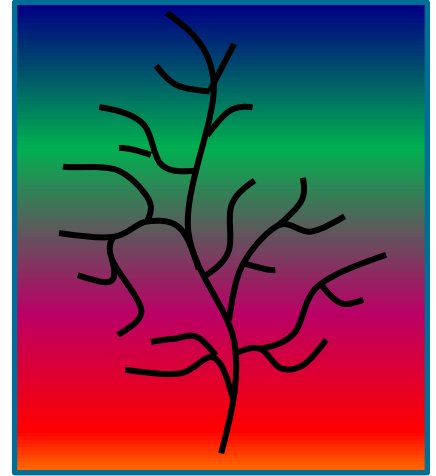


Figure 1 from Peterson et al. 2013

Taxonomy of Network Analyses

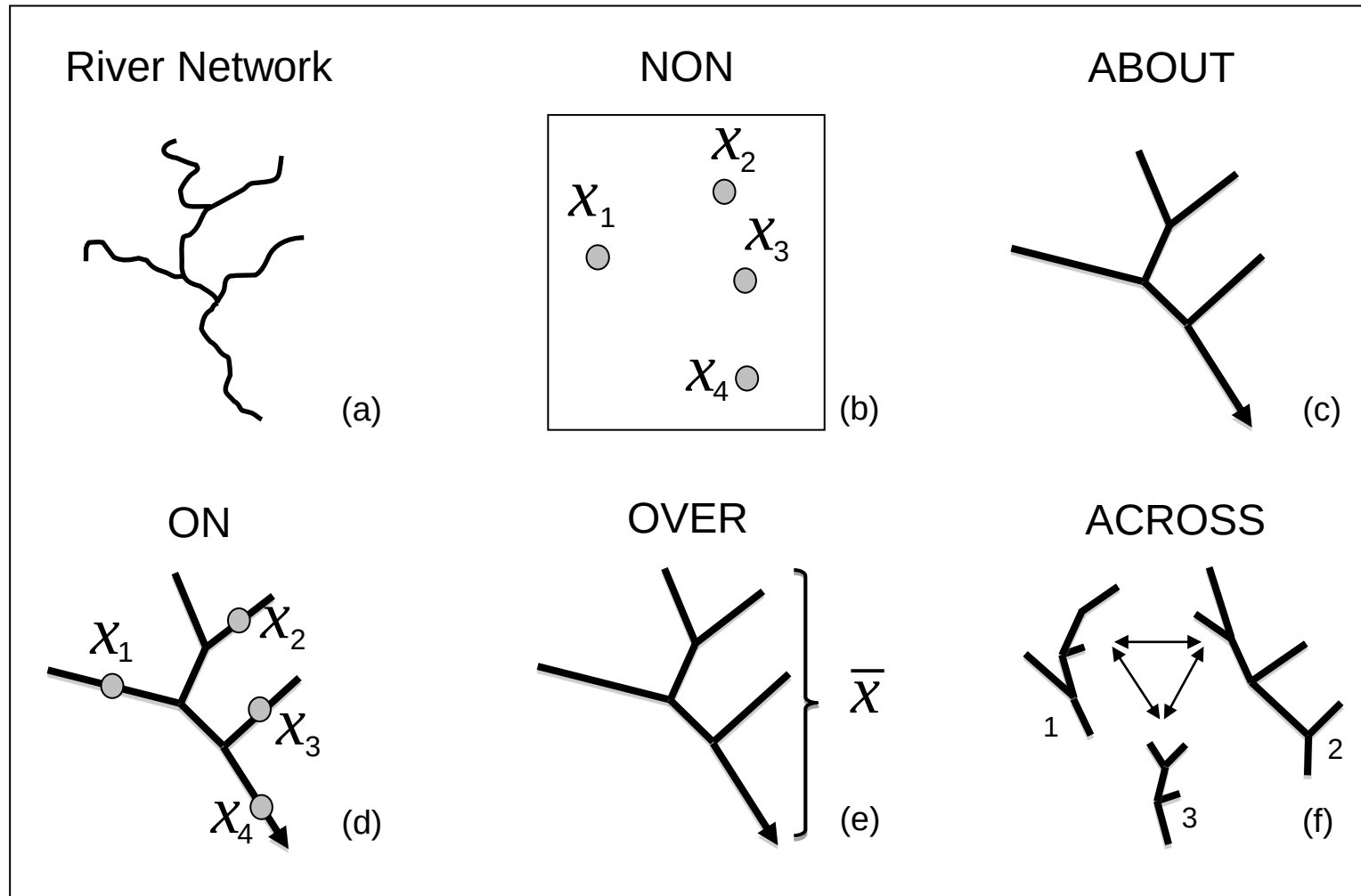


Figure 2 from Peterson et al. 2013

Taxonomy of Network Analyses

Purpose

1. Understand the differences between the processes of interest and the analytical methods available
2. Select the most suitable method for their study
3. Integrate network analyses to acquire a more coherent whole-of-system understanding

Taxonomy of Network Analyses

Range of data types used to describe the physical structure of streams, as well as the structure and function of ecological processes.

Fall into three general categories:

1. Physical structure
2. Biological or physicochemical processes
3. Aggregation of (1) or (2)

Data type limits the type of analysis you can use

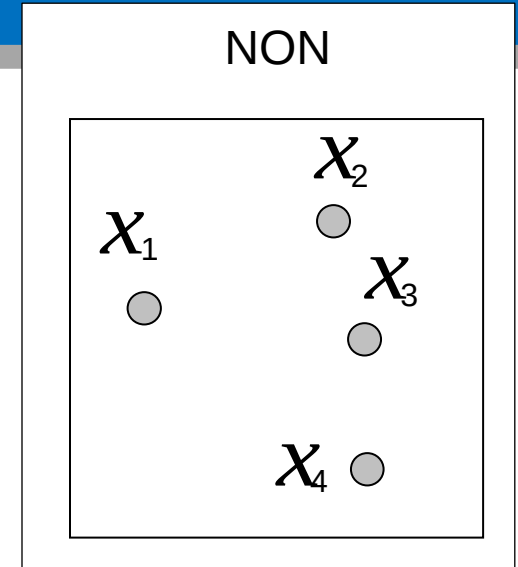
Table 1 from Peterson et al. 2013

Data type	Examples	Analysis method				
		NON	ABOUT	ON	OVER	ACROSS
Physical network structure (whole network)	Drainage density, fractal dimension or metric of network connectivity		✓			✓
Physical network structure (sub-network)	Confluence angle, stream order, node degree, edge weight, patch connectivity or composition	✓	✓	✓	✓	
Physicochemical and biological processes or attributes	Pool depth, pH or fish counts	✓		✓	✓	
Structure and processes aggregation	A special case where measurements are aggregated over an area (e.g. a hydrologic unit), a network, or multiple networks Mean fish count or confluence angle				✓	✓

Taxonomy of Network Analyses

OFF Network Analysis

- Ignore spatial relationships altogether
- Ignore network-based spatial relationships
- Use spatially explicit covariates to represent:
 - Sub-network structure, connectivity, or direction
 - Habitat quality, proximity, connectivity and arrangement
 - Least-cost path analyses and moving window approaches
 - Patch size, composition and distance measures
- Explore specific questions relating to the influence of physical network structure and in-stream habitat on physicochemical and biological stream processes

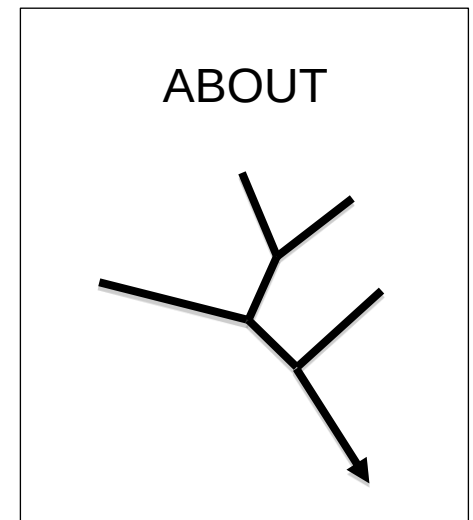


Difficult to complex, multiscale spatial processes and interactions within the dual coordinate system through spatially explicit covariates alone.

Taxonomy of Network Analyses

ABOUT Network Analysis

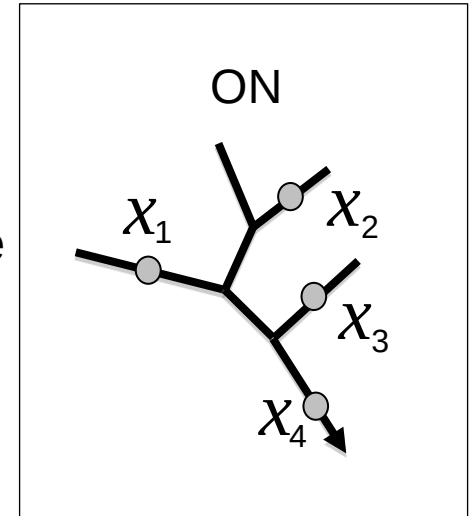
- Primary focus: physical network structure & connectivity of network itself
 - Whole-of-network: drainage density or fractal dimension
 - Sub-network: confluence angle, stream order
- Traditionally non-spatial – food webs at one location
- Spatially structured graphs
 - Modified to represent the dual coordinate system
- Temporally dynamic graphs: time-ordered networks
- Used to study the influence of structure on biological and physicochemical processes in streams
 - e.g. population structure, habitat connectivity, prioritization of conservation areas



Taxonomy of Network Analyses

ON Network Analysis (1):

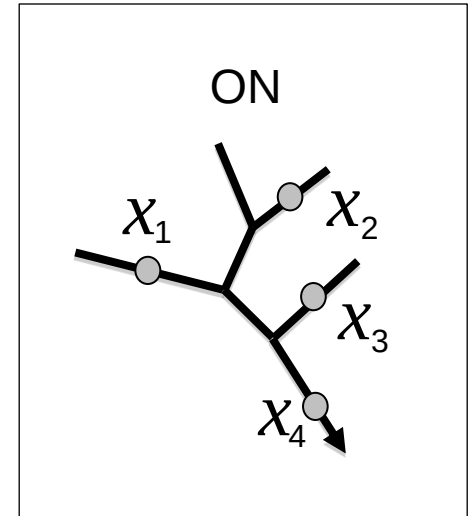
- Based on point data & sub-network structure
- Account for dual spatial representation
- Raster-based metapopulation models
- Typically used to investigate influence of network structure
 - fragmentation
 - movement behaviour of organisms
 - population distribution
 - metapopulation persistence
- Advanced theoretical understanding of network structure, connectivity, and function
- Not typically used to understand impacts of land management on in-stream habitat and organismal distributions
 - Lateral connectivity, in-stream movement is a function of distance, & in-stream habitat heterogeneity are usually ignored



Taxonomy of Network Analyses

ON Network Analysis: (2)

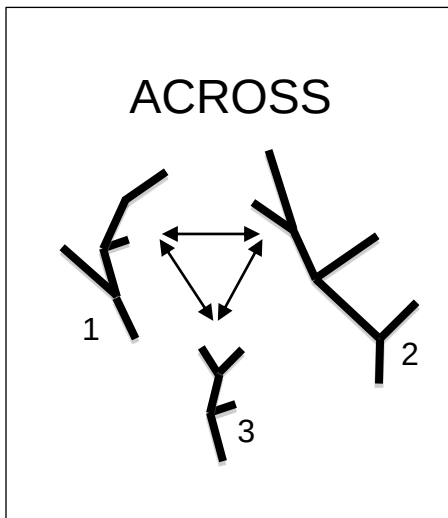
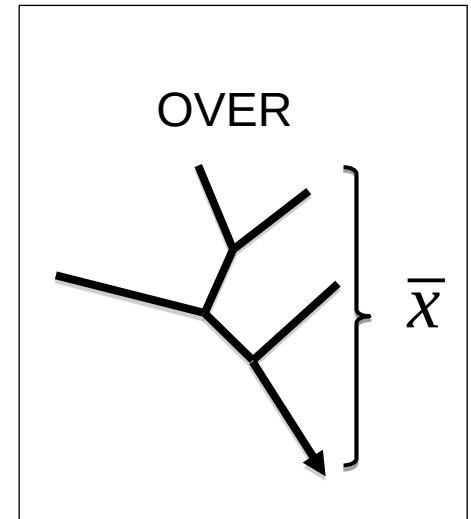
- Based on point data
- Account for dual spatial representation
- **Spatial statistical models *on* stream networks**
- Typically used to
 - Investigate influence of land-use on in-stream processes
 - Make predictions at unobserved locations, with estimates of uncertainty
- Not typically used to investigate the influence of network structure on stream processes



Taxonomy of Network Analyses

OVER Network Analysis

- Data: Physical network structure, physicochemical and biological processes or an aggregation of those characteristics
- Summarized *over* a network(s)



ACROSS Network Analysis

- Data: Whole-of-network structure or an aggregation of point data or sub-network structure
- Compare or contrast *across* networks or sets of networks

Taxonomy of Network Analyses

- **Each method and taxonomic class accounts for structure, connectivity, and directionality to some degree**
- **Choose the network-analysis most suitable for your question**
 - On or about-network models used to study the influence of physical network structure on biological processes
 - On- or over-network models used to study the effects of stream processes on another physicochemical or biological response
- **Integrate network analyses to address more complex questions**
 - Influence of structure and in-stream processes on species distributions
 - e.g. use a spatial statistical model to predict in-stream habitat conditions and then use as a restriction on movement in a metapopulation model

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